

PROJECT II

HR ANALYTICS

Employee Workforce, Performance & HR Actions Analysis

Project Detail	Value
Domain	Human Resources / People Analytics
Primary Tool	Microsoft Power BI
Database	MySQL
Data Preparation	Python / Power Query
Analysis	SQL + DAX
Dashboard Pages	4
Employee Records	1,562
Performance Records	9,605
HR Action Records	2,586

Final Project Report • September 2026

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Report structure

The report follows the actual project journey: raw HR data → cleaning → validation → modeling → SQL analysis → DAX → four Power BI dashboard pages → insights and recommendations.

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03. Project Introduction

This project is an end-to-end HR analytics solution built to convert raw employee, performance and HR-action data into a practical decision-support dashboard. It helps an HR or management team understand workforce size, employee composition, performance, tenure, HR activity and employee termination patterns.

Why HR Analytics?

HR analytics connects workforce data with business questions. Microsoft's Human Resources Power BI sample also focuses on new hires, active employees, former employees and separation trends, demonstrating how HR reporting can support workforce decisions.

Project Journey

Stage	What We Did	Output
1. Understand	Inspected tables, columns, IDs, dates and row counts.	Dataset understanding
2. Clean	Checked missing values, duplicates, types and invalid records.	Cleaner dataset

3. Validate	Checked dates, relationships and business rules.	Validated data
4. Model	Connected employee, performance and action tables.	Power BI model
5. Analyze	Performed SQL business analysis.	Business analysis
6. Calculate	Created DAX measures and calculated columns.	Dynamic KPIs
7. Visualize	Built four themed dashboard pages.	Interactive report
8. Interpret	Converted patterns into findings and recommendations.	Business insights
Project objective in one sentence Build a reliable HR dashboard that explains who the employees are, how they perform, how long they stay, what HR actions occur, and where termination patterns require attention.		

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04. Problem Statement & Objectives

Problem Statement

The original information was distributed across multiple HR tables. Looking at individual tables alone made it difficult to answer practical questions such as workforce size, active employees, tenure, performance, department patterns and termination risk.

The project therefore required more than visualization. It required a complete analytics process in which data quality, relationships, calculations and interpretation were handled in the correct order.

Project Objectives

Understand the structure and meaning of the employee, performance and action datasets.

Identify and resolve data-quality problems before using the data for business conclusions.

Build a clean and logically connected Power BI data model.

Use SQL to perform practical HR business analysis before visualization.

Create DAX measures for workforce, performance and termination KPIs.

Design four Power BI pages with clear business questions and varied visual types.

Identify workforce, performance and retention patterns.

Provide realistic recommendations while clearly stating limitations and assumptions.

Business Questions

How large is the workforce and how has it changed over time?

How are employees distributed by department, level, gender and race?

How long do employees stay with the organization?

How much of the workforce has performance information?

What is the average performance rating and how is performance distributed?

How do performance and tenure relate?

How many employees have terminated and what is the termination rate?

Which departments show higher termination counts?

What factors are associated with employee termination?

Interpretation rule

A dashboard can identify patterns and associations, but a pattern by itself does not prove that one factor causes employee termination.

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05. Data & Dataset Understanding

The project uses three main relational tables. The employee table represents employee master data, while performance and action tables contain multiple records associated with employees.

Main Tables

Table	Role	Key Fields / Examples	Granularity
tbl_employee	Employee master	EmpID, EmpName, EngDt, TermDt, DepID, GenderID, RaceID, MgrID, DOB, Level, PayRate	One row per employee
tbl_perf	Performance records	PerfID, EmpID, Rating,	Multiple records per performance date/year employee
tbl_action	HR action records	ActionID, EmpID, ActID, Year / date field	Multiple records per employee

Final Record Counts

Dataset	Final Rows	Purpose
Employee	1,562	Workforce population and employee attributes
Action	2,586	HR actions / employee events

Performance	9,605	Performance records and ratings
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Important Fields

EmpID — employee identifier and main relationship key.

EngDt — joining/engagement date used for tenure.

TermDt — termination date; blank TermDt is treated as active.

DepID — department identifier.

GenderID and RaceID — coded demographic attributes.

Level — employee/job level.

Rating — performance rating used for performance analysis.

ActionID / ActID — action-related identifiers; exact action-type meaning should be confirmed from the source or lookup table.

Data principle

The level of detail matters. Workforce count should use distinct EmpID rather than counting rows from performance or action tables, because one employee can have multiple records.

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06. Tools & Technologies

MySQL

Used for database creation, table loading, structural checks, validation queries and SQL business analysis.

Python

Used during early data understanding and quality checking: dimensions, data types, missing values, unique values and basic statistics.

Power Query

Used as a transformation and preparation layer before data reached the Power BI model.

Power BI

Used for the data model, DAX calculations, interactive visuals and the final four-page HR report.

DAX

Used for dynamic calculations such as employee count, active employees, terminated employees, average tenure, performance coverage and termination rate.

Tool	Main Use
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MySQL	Data storage, SQL queries, validation and business analysis
Python	Initial inspection, quality checks and descriptive analysis
Power Query	Data preparation and transformation
Power BI	Data model, dashboards and visual storytelling
DAX	Dynamic measures and calculated columns
Design approach	
Business questions were identified first; calculations and visual types were then selected to answer those questions. This reduces clutter and keeps the dashboard business-focused.	

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07. Project Workflow & Phases

The project followed a practical end-to-end analytics workflow. Each phase reduced a different risk: misunderstanding the data, poor data quality, incorrect relationships, incorrect calculations or misleading visuals.

Phase	Focus	Main Deliverable
Phase 1	Import & understand	Tables, columns, row counts, data types
Phase 2	Clean & prepare	Cleaned data and documented issues
Phase 3	Validate	Relationship, date and business-rule checks
Phase 4	Model	Employee-centered Power BI model
Phase 5	SQL analysis	HR business analysis queries
Phase 6	Power BI & DAX	Measures, calculated columns and visuals
Phase 7	Dashboard	Four-page interactive report
Phase 8	Review	Visual and logic corrections
Phase 9	Insights	Findings, recommendations and limitations

End-to-End Flow

RAW DATA

Employee + Performance + Action tables

- ↓ Data understanding and profiling
- ↓ Cleaning and validation
- ↓ MySQL relationships + SQL business analysis
- ↓ Power BI model + DAX
- ↓ Four dashboard pages
- ↓ KPI interpretation + business insights

FINAL OUTPUT

An interactive HR analytics report designed for workforce, performance and retention decision support.

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08. Phase 1 — Data Import & Initial Understanding

What We Did

Loaded employee, action and performance datasets into the database environment.

Checked tables and confirmed the expected datasets were present.

Checked row and column counts.

Reviewed data types such as integer, text and date fields.

Identified identifiers such as EmpID, PerfID and ActionID.

Reviewed joining and termination date fields.

Checked the level of detail in each table before building measures.

Dataset Snapshot

Table	Initial Purpose	Final Rows
tbl_employee	Employee master	1,562
tbl_action	Employee actions/events	2,586
tbl_perf	Performance records	9,605

Why This Phase Was Important

A common analytical mistake is to start building visuals before understanding the grain of the tables. Because performance and action tables contain multiple rows per employee, a simple row count could overstate headcount. Distinct employee counting was therefore established as the main workforce-counting logic.

Key learning

Before asking “What does the dashboard say?”, first ask “What exactly does one row represent?” This prevents many HR reporting errors.

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09. Phase 2 — Data Cleaning & Preparation

Major Data-Quality Issue: PayRate

The PayRate column was found to be empty across the employee dataset. Because compensation analysis requires reliable pay information, the project did not create salary-based conclusions from an empty field.

Decision made

Compensation/salary analysis was removed from the final project rather than filling missing values with assumptions. This keeps the dashboard evidence-based.

Cleaning Activities

Checked missing values in important fields.

Checked duplicate employee records.

Reviewed date columns and formats.

Checked numeric columns and basic ranges.

Reviewed coded category fields such as DepID, GenderID and RaceID.

Prepared fields for Power BI analysis.

Created the Tenure Years calculation using joining date and termination/current date.

Tenure Calculation

For active employees, the current date is used as the end date. For terminated employees, TermDt is used.

DAX

```
Tenure Years = DATEDIFF(tbl_employee[EngDt], COALESCE(tbl_employee[TermDt], TODAY()), DAY) / 365.25
```

Cleaning Principle

If a business field cannot support reliable analysis, it is better to exclude that analysis and document the limitation than create artificial values simply to complete a dashboard.

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10. Phase 3 — Data Validation & Quality Checks

Business-Rule Checks

During validation, the project identified records that required investigation against employee dates.

Validation Check	Records Found	Interpretation
Action before employee joining date	104	Potential date/data-quality issue requiring review
Action after employee termination	11	Potential date/data-quality issue requiring review

These records were treated as validation findings rather than automatically being interpreted as real business events. Date anomalies can arise from data-entry timing, historical records or mismatched event dates.

Other Validation Checks

Employee IDs were checked as the main relationship key.

Performance and action records were checked against the employee population.

Required employee fields were reviewed.

Date fields were reviewed for logical consistency.

Duplicate and uniqueness checks were performed on employee identifiers.

Performance ratings were reviewed for expected values.

Numeric fields were reviewed for invalid or unexpected values.

Validation lesson

Data quality is part of the analysis. A visually attractive dashboard built on invalid relationships or dates can still produce wrong business conclusions.

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11. Phase 4 — Data Modeling

Model Structure

The employee table acts as the central employee-level table. Performance and action records are linked through EmpID. This allows employee attributes such as department, level and demographics to filter related performance and action records.

Relationship concept

tbl_employee[EmpID] → tbl_perf[EmpID]

tbl_employee[EmpID] → tbl_action[EmpID]

Core Workforce Measure

DAX

Employee Count = DISTINCTCOUNT(tbl_employee[EmpID])

Why DISTINCTCOUNT?

The employee table contains one row per employee, but action and performance tables can contain multiple records for the same person. DISTINCTCOUNT prevents repeated employee records from being treated as additional employees.

Employee Status Logic

DAX

```
Employee Status = IF(ISBLANK(tbl_employee[TermDt]), "Active", "Terminated")
```

Modeling Outcome

Employee attributes can filter performance and action analysis.

Measures can be reused across dashboard pages.

Active and terminated workforce can be compared consistently.

Tenure can be calculated once and reused in multiple visuals.

The model supports interactive filtering instead of fixed static numbers.

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12. Phase 5 — SQL Business Analysis

SQL analysis was used before dashboard construction to test business logic and understand patterns directly from the database. This reduced the risk of creating visuals that looked correct but used incorrect logic.

Main SQL Analysis Areas

Area	Business Question	Typical Output
Workforce	How many employees are there?	Distinct employee count
Department	Where are employees concentrated?	Employees by DepID
Status	How many are active vs terminated?	Status counts
Tenure	How long do employees stay?	Average / distribution
Performance	What are the rating patterns?	Rating distribution / average
Performance by department	Which departments differ?	Average rating by DepID
Actions	How active is HR movement?	Action counts / trends
Termination	Where and when do exits occur?	Department/year counts

Example: Employees by Department

SQL

```
SELECT DepID, COUNT(*) AS Emp_Dep
FROM tbl_employee
GROUP BY DepID
ORDER BY DepID ASC;
```

Business Logic Used

TermDt blank → Active employee.

TermDt populated → Terminated employee.

Employee counts → DISTINCT EmpID where multiple-row tables are involved.

Average tenure → employee-level tenure rather than action-row tenure.

Performance → rating-based analysis from tbl_perf.

Department termination counts → useful for volume, but not equivalent to department turnover rate.

Important distinction

A department with more terminated employees may simply be larger. Proper department turnover comparison should normalize terminations by department workforce size.

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13. Phase 6 — Power BI & DAX

Core Measures

Measure	Logic / Purpose
Employee Count	DISTINCTCOUNT(tbl_employee[EmpID])
Active Employees	Distinct employees where TermDt is blank
Terminated Employees	Distinct employees where TermDt is not blank
Average Tenure	AVERAGE(tbl_employee[Tenure Years])
Average Rating	AVERAGE(tbl_perf[Rating])
Performance Coverage %	Employees with performance / total employees
Termination Rate %	Terminated employees / total employees
Avg Tenure at Exit	Average tenure filtered to terminated employees

Final Dashboard KPIs

Performance Coverage

DIVIDE([Employees With Performance], [Employee Count], 0)
Final dashboard value: 98.08%

Termination Rate

DIVIDE([Terminated Employees], [Employee Count], 0)

Final dashboard value: 28.43%

Avg Tenure at Exit

Average tenure filtered to terminated employees

Final dashboard value: 4.4 years

Why DAX Matters

DAX measures are dynamic and respond to filters and visual context. HR users can therefore examine the same KPI by department, level, year or other dimensions without creating separate static calculations.

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14. Page I — Workforce Overview

Purpose

Page I provides the high-level workforce picture. It answers: “What does the workforce look like overall?” The page establishes the baseline before moving into employee, performance and termination detail.

Visual Story

Visual	Type	Purpose
Total Employees	KPI/Card	Overall workforce size
New Hires	KPI/Card	Hiring activity
Attrition / Termination Rate	KPI/Card	Workforce movement
Headcount by Department	Bar	Department concentration
Headcount by Gender	Donut/Column	Gender composition
Headcount by Level	Bar	Level distribution
Headcount Trend	Line	Workforce movement over time
Active vs Terminated	Column	Overall workforce status

Business Interpretation

The workforce baseline is 1,562 employees.

Department and level distributions show where the workforce is concentrated.

Active vs terminated status provides immediate context for workforce stability.

The time trend helps identify periods of workforce growth or contraction.

Demographic visuals are descriptive and should not be treated as proof of fairness or bias without additional analysis.

Design

Page I uses a Sky Blue theme. KPIs come first, followed by composition and trend visuals.

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15. Page II — Employee Analysis

Purpose

Page II moves from the overall workforce picture into employee-level composition and stability. It focuses on growth, employee level, demographics, department, performance coverage and tenure.

Visual	Type	Business Question
Employee Growth Over Time	Line/Column	How has employee count changed?
Employees by Level	Column	How is the workforce distributed by level?
Workforce by Gender	Donut	What is the gender composition?
Employees with Performance Records	KPI	How many employees have performance data?
Average Employee Level	KPI	What is the average level?
Average Employee Tenure	KPI	How long do employees stay?
Workforce by Race	Donut/Column	What is the workforce composition by race?
Employees by Department	Bar	Which departments are larger?
Tenure by Employee Level	Bar/Column	How does tenure vary by level?
Active Employees	KPI	How many employees are active?

Key Interpretation

Employee distribution is visible across departments and levels.

Tenure provides a stability perspective that headcount alone cannot provide.

Performance coverage is high, allowing the performance page to represent most of the workforce.

Demographic breakdowns provide descriptive workforce composition.

Level and tenure together can help identify progression and retention patterns.

Visual design

Page II uses a Dark/Sky Blue identity and combines KPI cards, bars, a donut and tenure analysis.

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16. Page III — Performance Analysis & Insights

Purpose

Page III answers: “How are employees performing, and is performance related to tenure or workforce characteristics?”

Visual	Type	Role / Finding
Performance Distribution	Clustered Column	Shows rating frequency; rating 3 is the central/most common level in the observed data.
Performance Rating Mix	Donut	Shows rating composition.
Average Rating by Department	Bar	Compares department-level average rating.
Performance Trend	Line	Shows average rating over time.
Average Rating	KPI	3.01 overall average rating.
High Performers	KPI	1,239 under the project definition.
Performance Coverage	KPI	98.08% of employees have performance records.
Performance vs Tenure	Scatter	Tests visible relationship between tenure and rating.
Average Rating by Gender	Column	Compares average rating across gender groups.

Key Insights

Overall average performance rating is approximately 3.01.

Performance coverage is very high at 98.08%.

High performers total 1,239 under the project definition.

The performance-vs-tenure scatter does not show a strong obvious positive relationship.

Department and gender comparisons show relatively limited differences in average rating in the observed data.

Performance distribution should be read together with the average rating.

Important definition

“High Performer” is a project definition and should not be treated as a universal HR standard.

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17. Page IV — HR Actions & Insights

Purpose

Page IV is the retention and workforce-movement page. It focuses on HR actions, termination patterns, workforce status and factors associated with employee termination.

Visual	Type	Role
Employee Actions by Type	Donut	Distribution of action-code categories; labels require verified lookup.
Actions Over Time	Line	Tracks annual HR action volume.
Terminated Employees by Department	Bar	Ranks termination counts by department.
Employee Termination Trend	Area	Shows termination counts over time.
Average Tenure at Exit	KPI	4.4 years.
Workforce Status by Department	100% Stacked Column	Compares active vs terminated composition.
Active vs Terminated Employees	Treemap	Overall workforce-status proportion.
Employee Termination Rate	KPI	28.43%.
Factors Influencing Employee Termination	Key Influencers	Explores statistical associations with termination.

Observed Findings

Overall termination rate is 28.43% using the project’s defined calculation.

Average tenure at exit is 4.4 years.

Department termination counts range approximately from 37 to 53 in the observed dashboard.

The termination trend rises toward 2023 and drops sharply in 2024; completeness should be checked before calling this an improvement.

Key Influencers is configured to analyze Employee Status = Terminated and uses DepID, Level, GenderID, RaceID and Tenure Years.

Key Influencers results are associations, not proof of cause.

Action-code caution

ActionID is normally a record identifier, while ActID may represent an action category/code. Do not label codes such as 10/30/90/91 as promotion, termination, etc. unless the source confirms those meanings.

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18. Key Findings, Recommendations, Limitations & Conclusion

Top 10 Findings

Workforce size: 1,562 employees are represented in the final employee dataset.

Performance coverage: 98.08% of employees have performance records.

Average performance: overall average rating is approximately 3.01.

High performers: 1,239 employees meet the project's high-performer definition.

Termination rate: 28.43% of the employee population is classified as terminated under the project measure.

Average tenure at exit: departing employees average about 4.4 years of tenure.

Department termination volume: observed counts range roughly from 37 to 53.

Performance vs tenure: the scatter does not indicate a strong simple relationship.

Termination trend: the series rises toward 2023 and drops sharply in 2024; completeness should be verified.

Data quality matters: missing PayRate and date anomalies demonstrate why validation must happen before interpretation.

Recommendations

Calculate department-level termination rates, not only termination counts.

Investigate exits around the observed mid-tenure period, including career progression and manager experience.

Add reliable compensation data so pay equity and compensation-retention analysis can be performed.

Add structured termination reasons and voluntary/involuntary separation fields.

Add engagement, absenteeism, promotion and training data for deeper retention analysis.

Create a recurring HR data-quality process for dates, missing values and relationships.

Continue monitoring the dashboard monthly or quarterly and investigate significant changes.

Limitations

PayRate is missing, so compensation analysis was excluded.

Some categories are represented by coded IDs, limiting direct business interpretation without lookup tables.

Termination counts are not the same as normalized turnover rates by department.

The 2024 termination decline may be affected by incomplete or partial-year data.

Key Influencers identifies associations; it does not establish causation.

The project is descriptive and diagnostic; future work could add predictive attrition modeling.

Conclusion

This project demonstrates a complete analytics workflow from raw HR data to an interactive Power BI report. The strongest part of the project is the sequence of understanding, cleaning, validating, modeling, analyzing and then visualizing the data. The final dashboard provides a practical foundation for workforce planning, performance review and retention analysis.

Final takeaway

Reliable HR analytics requires reliable data, clear definitions and disciplined interpretation. The dashboard should support HR decisions alongside organizational context, not replace professional judgment.

Selected References

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